Evaluating Domain Adaptation Techniques for Classical Machine Learning Models

Research paper draft for faculty review

Abstract

Distribution shift can reduce the reliability of classical classifiers. This study evaluates domain-classifier importance weighting and centred, regularised CORAL against source-only Logistic Regression, linear Support Vector Machine and Random Forest models. Experiments use UCI Gas Sensor Array Drift, smartphone Human Activity Recognition and the original four-domain Amazon sentiment corpus, covering 22 transfers and five seeds. Source-only validation selects shared hyperparameters; separate target adaptation features support adaptation, while held-out target observations remain excluded until evaluation. The completed study contains 1,100 primary and 180 sensitivity evaluations. Across 132 primary adaptation comparisons, mean macro F1 improves in 40 and declines in 92; 27 descriptive group-bootstrap intervals include zero. CORAL benefits several gas transfers but adaptation generally harms HAR and Amazon performance. Accuracy, runtime, weight concentration and covariance diagnostics contextualise these outcomes. Shared test observations and few independent domains limit inference. The contribution is reproducible comparative evidence and an independently auditable local research workbench.

1 Introduction

Classical classifiers are useful when labeled data and computing resources are limited, but a change between training and deployment inputs can undermine their reliability. Adaptation may exploit unlabeled target observations, although improvement depends on assumptions about support and label relationships. This study asks whether importance weighting and covariance alignment help LR, linear SVM and RF under three distinct applications. It reports negative transfer and computational cost rather than selecting a method only from successful examples.

The frozen objective is:

To evaluate whether importance weighting and feature alignment using CORAL improve the performance of Logistic Regression, Support Vector Machine, and Random Forest when source and target data follow different distributions. The study will compare these techniques against models trained without adaptation across three public datasets, using accuracy, macro F1 score, and computational time. It will identify the conditions under which adaptation improves performance, provides little benefit, or reduces prediction quality.

The contribution is a controlled, reproducible comparison of established methods and an application that exposes the exact evidence behind each displayed result. No new adaptation algorithm or publication acceptance is claimed.

2 Related work

Discriminative estimation links domain-membership probabilities to density ratios [R11]. Importance-weighted validation [R02] and kernel mean matching [R03] are related approaches, but neither is implemented here. CORAL aligns covariance statistics [R01], whereas subspace alignment uses eigenvector mappings [R06]. Adaptation theory includes labeling compatibility as well as distribution divergence [R04]; reduced marginal discrepancy alone cannot guarantee target accuracy. Transfer-learning terminology and negative transfer are reviewed in R07. Dataset context comes from the original Amazon sentiment [R05], gas drift [R08] and smartphone HAR [R09] papers. Their training protocols differ from ours, so published scores are not directly compared with this study.

3 Datasets and experimental protocol

Gas Sensor Array Drift uses 13,910 retained measurements with 128 features and nine transfers from Batch 1 to Batches 2-10. Each target batch is divided into separate adaptation and evaluation groups using split seed 2026. HAR uses 10,299 retained windows and 561 features. Its source contains 21 official training participants; four official test participants form adaptation and the other five form evaluation. Amazon uses the original four categories, including the separate books unlabeled file, with all twelve ordered transfers. Deduplication retains 70,509 of 78,679 records; only review bodies enter the text model.

Training uses source task labels and target adaptation features. Target evaluation features and labels are unavailable to preprocessing, model selection, weighting and covariance estimation. A completed-fit lock authorizes the final evaluator. Source scalers and, for Amazon, TF-IDF followed by at most 128 SVD components are refitted inside each source CV fold. All methods share the permitted representation and source-selected base hyperparameters. HAR folds preserve whole participants.

LR and LinearSVC search C in {0.1, 1, 10}. RF uses 200 trees and four combinations of depth unlimited or 20 and minimum leaf size 1 or 3. Five source folds optimize macro F1. The study uses seeds 17, 29, 43, 71 and 101, a source-majority baseline, and a single CPU job with one numerical thread. Scikit-learn 1.9.1 [R10, S01-S06] is installed within the actual 64-package lock.

4 Adaptation and analysis

Importance weighting fits a logistic domain classifier with C 1 and five source cross-fitting folds. For target-membership probability q, source fold-training count ns and adaptation count nt, the ratio is `q/(1-q) * ns/nt`. Probabilities are bounded to [0.001, 0.999], ratios to [0.05, 10], then weights are normalized to source mean one. All task classifiers receive sample weights. ESS measures weight concentration and warnings are retained.

The centred CORAL implementation adds ridge 0.001 to unbiased source and target adaptation covariances, computes `A = Cs^(-1/2) Ct^(1/2)` by symmetric eigendecomposition with floor 1e-8, and trains on `(Xs-ms)A+mt`. Only the source features receive this alignment transform. Mean restoration and the chosen ridge are explicitly disclosed implementation choices.

Metrics are calculated from saved predictions. Macro F1 uses the full target task's present classes with zero_division 0. Paired deltas share evaluation rows, base parameters and seeds. Descriptive 95% percentile intervals use 1,000 paired group resamples, averaging deltas over five fitted seeds without refitting. Groups are HAR participants, Amazon products where available and duplicate-safe gas measurements. These unadjusted intervals describe conditional test-sample uncertainty, not independent replications across seeds.

5 Results

All 1,100 primary and 180 additional sensitivity evaluations completed. Thirty source searches generated 1,500 candidate-fold scores. Among 132 primary adaptation comparisons, 40 mean macro-F1 changes are positive and 92 negative; 27 corresponding intervals include zero. There were zero classifier fitting failures. Two operational progress-file errors were retained and resumed.

Dataset	Method	Pairs	Positive	Negative	CI includes 0
amazon	CORAL	36	8	28	7
amazon	Weighting	36	3	33	4
gas	CORAL	27	19	8	5
gas	Weighting	27	10	17	9
har	CORAL	3	0	3	1
har	Weighting	3	0	3	1

Table 1 counts task-model comparisons within each dataset and adaptation method. Tasks share data and these counts are descriptive. CORAL's mean gas improvements for LR and linear SVM are 22.576452 and 11.347881 percentage points across nine transfers. All six primary HAR adaptations reduce mean macro F1, and most Amazon comparisons also decline. Amazon RF with CORAL has a small positive dataset mean, illustrating why task effects must accompany an overall ranking.

Example	Transfer	Model	Method	Delta pp	95% interval pp
positive	Gas 1→07	LR	CORAL	+41.284465	[+38.973957, +43.482547]
uncertain	Gas 1→08	Linear SVM	Weighting	-0.148662	[-1.266615, +0.828300]
negative	Gas 1→02	RF	CORAL	-34.075518	[-36.618668, -18.709011]

Table 2 uses saved five-seed paired macro-F1 means and group-bootstrap intervals in percentage points. The examples represent a gain, an uncertain small loss and a large loss. Their exact source/adapted run IDs appear in the accompanying claim-trace file and complete paired table.

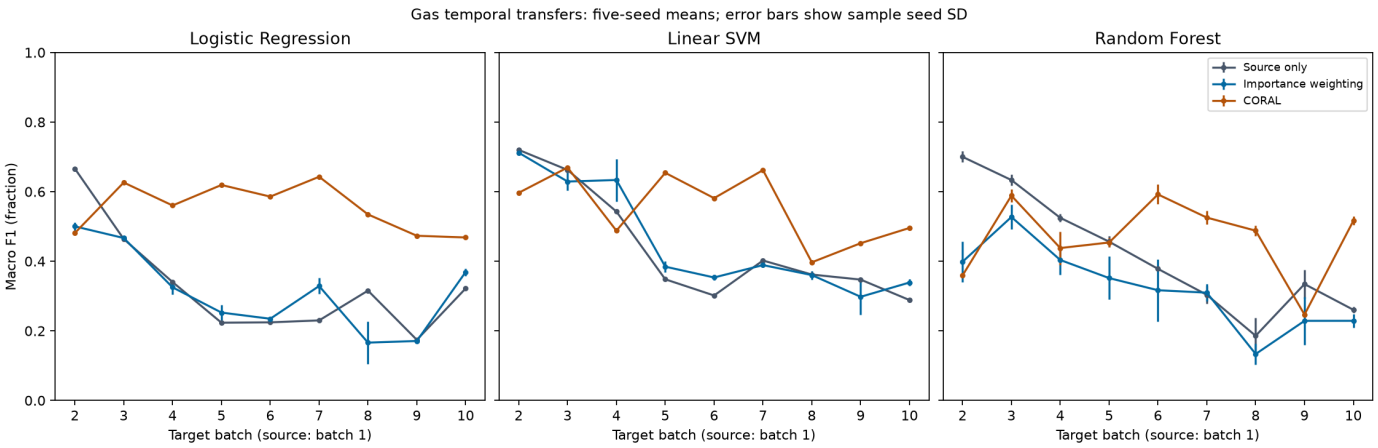


Figure 1. Gas Batch 1 to Batches 2-10, macro F1 in fractions. Points show primary five-seed means; error bars show sample seed SD. Tasks share the source batch. The saved figure-trace manifest identifies the exact runs behind every series.

Computational time includes preprocessing, adaptation, classifier fit and target prediction, with source search reported separately. Representative gas, HAR and Amazon benchmarks measured peak process working sets of approximately 284, 687 and 517 MiB. The benchmark-derived full-study planning estimate was 2.98-3.97 hours, not a measured total elapsed runtime. Cached component costs are retained and must not be summed repeatedly as unique study computation.

Sensitivity retains clipping 5, 10 and 20 and ridge 0.0001, 0.001 and 0.01 on the three declared diagnostic tasks. Gas linear SVM macro F1 increases from 0.319973 to 0.354481 between the extreme clipping settings, while HAR RF decreases from 0.934667 to 0.923518. Primary settings remain fixed after seeing these results. Lower covariance mismatch and larger ESS are diagnostics rather than automatic certificates of better classification.

6 Reproducibility and limitations

A clean checkout and fresh virtual environment reproduced gas Batch 1 to Batch 2 using all five seeds and 50 newly fitted models/baselines. All 50 ordered prediction vectors, metric dictionaries and selected parameters exactly matched the original study on 622 evaluation rows per run. Fitting took 204.295694 seconds; fitting, evaluation and comparison took 245.421423 seconds. Peak process working set was 216.0625 MiB. No forbidden held-out training reads occurred. The original archive/split were verified and reused; new caches/models/predictions were written separately. The detailed reproduction record preserves the exact executed driver and logs.

The source package provides immutable split identities, configurations, exact versions, checksummed traces, complete result tables and real dashboard screenshots. Raw archives and serialized models are excluded from distribution. Acquisition instructions explain the documented source terms. Fresh-environment reproduction uses verified local data and the same hardware/OS, so it does not establish portability across all platforms.

Source-only tuning may differ from target-optimal selection. Density-ratio support and stable conditional-label assumptions may fail; CORAL does not align all class structure. Amazon text uses a deliberately bounded common representation. There are only three applications and five HAR evaluation participants. Shared test data, overlapping Amazon tasks, possible within-batch gas dependence, varying target-present class sets and unadjusted intervals limit inference.

7 Conclusion

Adaptation produced substantial gains in some gas transfers but was often ineffective or harmful elsewhere. The evidence favors evaluating a matched source-only baseline, preserving negative transfer and inspecting assumptions and computational cost. It does not support a universal adaptation winner. Future independent-domain or target-budget studies require a new declared protocol, rather than post hoc changes to these primary settings.

Assistance and submission status

Codex supplied substantial implementation, experiment execution, checks, writing and document-review assistance, as detailed in the companion contribution record. The student's independent review and viva rehearsal are not recorded as completed. This is a neutral academic draft for faculty review; no target venue, author approval, institutional endorsement or publication status has been invented.

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Checked against the primary publisher, proceedings, author or official software source on 11 October 2026. Method and dataset claims below are attributed to their original sources. Project scores come from the saved local study, not these papers.

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Verification scope

R01-R05, R07, R09-R11 were checked using primary full-paper text, including the method or dataset sections used here. R06 was checked using the indexed CVF proceedings abstract and metadata; direct HTML/PDF access returned HTTP 403. R08 was checked using the indexed publisher abstract, introduction and method preview; direct publisher access also returned HTTP 403. No unseen experimental detail from either restricted paper is claimed. IEEE's page required JavaScript; the author-hosted R07 manuscript supplied the text. The DOI, volume and page metadata identify the final publication separately from that manuscript.

D01-D03 and S01-S06 were opened directly. The online scikit-learn documentation identified itself as 1.9.1, matching the installed scientific environment. The reading matrix records limitations and relevance; it does not claim that the student independently read or verified every source.